On commensurable pro-$\mathcal{C}$ groups

Pedro Cusinato Loiola Lima
Universidade de Brasília

Let $\mathcal{C}$ be an extension-closed variety of finite groups. Two pro-$\mathcal{C}$ groups are said to be commensurable if they share a common open subgroup. In this talk we give necessary and sufficient conditions to embed commensurable pro-$\mathcal{C}$ as open subgroups of a common overgroup. In the abstract setting, this problem was recently studied by Minasyian and Touikan [1]. We also consider the analogous problem for residually finite groups.

Keywords: profinte groups, residually finite groups

References

[1] N. Touikan, A. Minasyan, On the proofs of Leighton's graph covering theorem, a notion dual to commensurability, and normal virtual retracts. *Preprint* disponível em https://arxiv.org/abs/2507.01839(2025).